

Problem 15.13

Over the next 20 years, you deposit money into a retirement account at the end of each year according to the following schedule:

Time	Amount invested each year
1 – 5	\$2,000
6 – 10	\$3,000
11 – 20	\$5,000

The effective annual rate of interest is 9%. Find the accumulated value of your account at time 20.

Solution

The annual effective interest rate is

$$i = 0.09.$$

We accumulate each block of payments to time 20.

Deposits from years 1–5

The first five deposits form an annuity-immediate of \$2,000 accumulated to time 5, then accumulated for another 15 years to time 20. Its value at time 20 is

$$2000s_{\overline{5}|} (1.09)^{15}.$$

Deposits from years 6–10

The next five deposits form an annuity-immediate of \$3,000 accumulated to time 10, then accumulated for another 10 years to time 20. Its value at time 20 is

$$3000s_{\overline{5}|} (1.09)^{10}.$$

Deposits from years 11–20

The final ten deposits form an annuity-immediate of \$5,000 accumulated to time 20. Its value at time 20 is

$$5000s_{\overline{10}|}.$$

Therefore, the total accumulated value is

$$AV = 2000s_{\overline{5}|} (1.09)^{15} + 3000s_{\overline{5}|} (1.09)^{10} + 5000s_{\overline{10}|}.$$

Using

$$s_{\overline{n}|} = \frac{(1+i)^n - 1}{i},$$

we get

$$\begin{aligned} AV &= 2000 \left(\frac{(1.09)^5 - 1}{0.09} \right) (1.09)^{15} \\ &\quad + 3000 \left(\frac{(1.09)^5 - 1}{0.09} \right) (1.09)^{10} \\ &\quad + 5000 \left(\frac{(1.09)^{10} - 1}{0.09} \right). \end{aligned}$$

Finally,

$$AV \approx 162067.0149.$$

Hence, the accumulated value of the account at time 20 is

$$\boxed{\$162,067.01}.$$